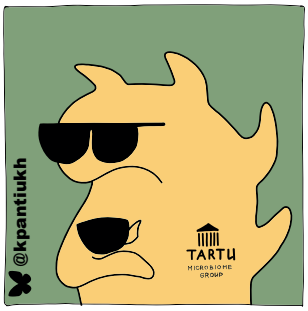

Species structure of human gut microbiome species

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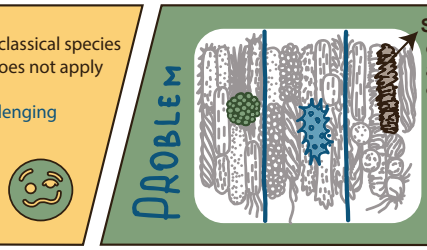
Tartu Microbiome group

Bacteria do not reproduce sexually, so the classical species concept based on reproductive isolation does not apply

Defining bacterial species is therefore challenging

Genomes within a species often form a **continuum of genetic similarity** rather than distinct clusters

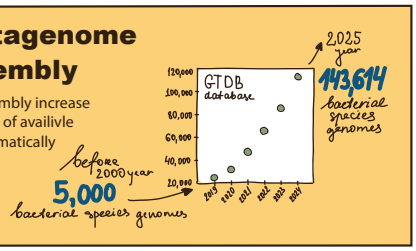


Species representative
one bacterial species - a group with more then 95% genome similaity

species 1
species 2
species 3

Metagenome assembly

MAG assembly increase the number of availivle genomes dramatically



QUESTION: Do gut bacterial species share common patterns of subspecies structure?

RESULTS

General trends

Is the hypothesis of a species continuum valid for gut bacteria?
Or do genomes form distinct clusters of highly similar units?
If clustering occurs, what ANI threshold defines these groups?

To investigate, we analyzed the distribution of within-species pairwise ANI values across all genomes of the 376 selected species
RESULTS: A total of **6,942,981** comparison (ANI*) values
*-Average Nucleotide Identity

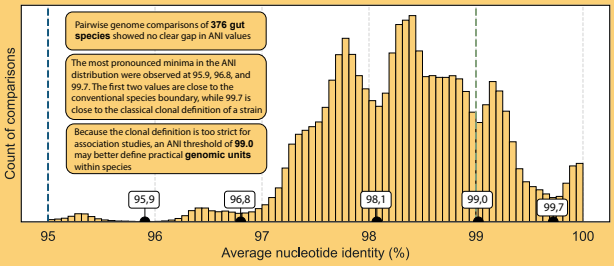


FIGURE 1. Histogram of within-species pairwise genome comparisons for 376 gut species, each represented by more than 100 MAGs reconstructed in the EstMB dataset.

RESULTS

Species-specific structure

Example 1. Species with large, well-defined subspecies clusters

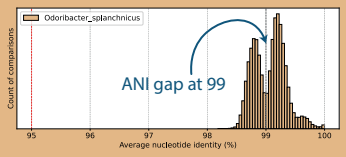


FIGURE 2. Histogram of within-species pairwise genome comparisons for *Odoribacter splanchnicus*

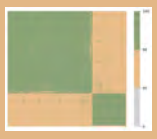


FIGURE 3. Heatmap of within-species pairwise genome comparisons for *Odoribacter splanchnicus*

FIGURE 4. Specie structure of *Odoribacter splanchnicus*

Example 2. Species without subspecies clusters

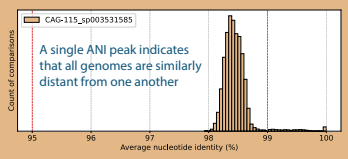


FIGURE 5. Histogram of within-species pairwise genome comparisons for CAG-115 sp003531585

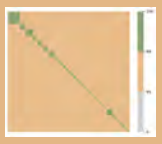


FIGURE 6. Heatmap of within-species pairwise genome comparisons for CAG-115 sp003531585

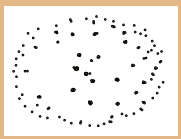


FIGURE 7. Specie structure of CAG-115 sp003531585

Example 3. Complex hierarchical subspecies structure

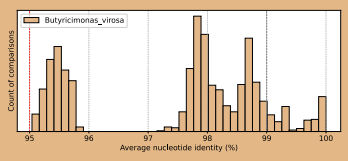


FIGURE 8. Histogram of within-species pairwise genome comparisons for *Butyricimonas virosa*

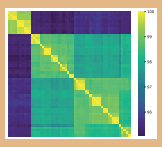


FIGURE 9. Heatmap of within-species pairwise genome comparisons for *Butyricimonas virosa*

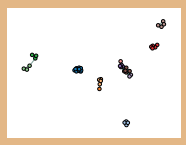


FIGURE 10. Specie structure of *Butyricimonas virosa*

METHODS

Sequencing: shotgun paired-end 150 bp reads with deep coverage (56M paired reads/sample on average).

MAGs Assembly: Multi-binning metagenome assembly (VAMB, MaxBin, MetaBAT), refined with DAS-Tools. MAG quality was assessed using CheckM2, GUNC, and SeqKit. MAGs were cluster with dRep. Taxonomic annotation was performed with GTDB-Tk.

WHY IMPORTANT?

Understanding subspecies structure is critical for effective microbiome-wide association studies.

Without it, WMAS treats diverse strains as a single unit, which reduces statistical power and hides real biological signals.

PREPRINT

check pre-print about MWAS results here

CONCLUSIONS

- Overall, no universal ANI gap exists across gut bacterial species. The only consistent distribution minimum is weak and occurs around ANI = 99 (FIgure 1).
- Gut species display highly diverse sub-species organization. Some species form large, well-defined subspecies clusters, others show no clear clustering at all, and many exhibit complex hierarchical structures.
- Even among species with similar patterns of organization, the ANI thresholds that define clusters vary widely, indicating that no single cutoff can be applied universally.

Next - new approach

for sub-species analysis

Analise within-species pairwise genome comparisons distribution & define ANI threshold for sub-specie **Genome Units** (GU)
Calculate the number of GU per species - GUN

$$\text{normGUN} = \frac{\text{GUN}}{\text{number of MAGs}}$$

This project was supported by:

European Union European Regional Development Fund
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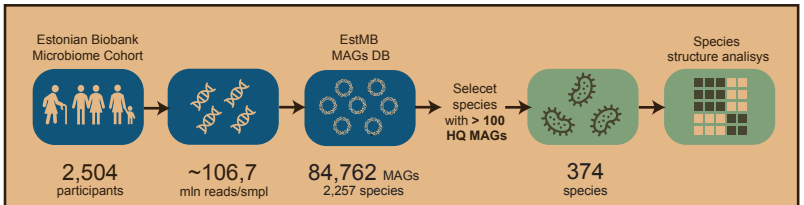
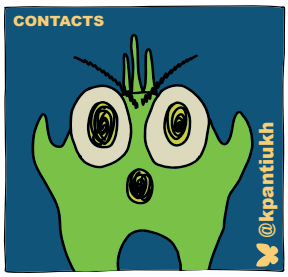
Pipeline visualisation